

# Upskilling I Linux e Segurança na Cloud

## Administração para Linux em Cloud

### Sessão 1 | Atividade prática individual 1

**Mini-servidor Linux funcional, seguro e documentado**

# Relatório do Trabalho Prático individual 1

## Preparação do ambiente Linux e organização da primeira evidência técnica

### Objetivos da atividade

- identificar o ambiente *Linux* utilizado;
  - distinguir *VM* local, *VPS*, infraestrutura em nuvem e *Linux* no *browser*;
  - executar comandos iniciais no terminal;
  - criar uma estrutura de pastas para organizar evidências;
  - registar comandos executados e a sua finalidade;
  - iniciar um pequeno *README* técnico.
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### Tarefa 1 - Preparar ou aceder ao ambiente Linux

- **Trilho seguido:** Trilho B — Linux no browser
  - **Ambiente utilizado:** Linux no browser
  - **Estado do ambiente:** Funcional
  - **Dificuldade encontrada:** Algumas dificuldades iniciais com comandos Linux
  - **Plano para continuar:** Continuar a praticar comandos básicos e organização de ficheiros
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### Tarefa 2 - Criar a estrutura inicial do projeto

Estrutura de diretórios e ficheiros:

```
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root@ubuntu:~$ mkdir -p linux-seguranca-cloud/topico-01/evidencias
root@ubuntu:~$ mkdir -p linux-seguranca-cloud/produto-final
root@ubuntu:~$ touch linux-seguranca-cloud/topico-01/comandos-topico-01.txt
root@ubuntu:~$ touch linux-seguranca-cloud/topico-01/README.md
root@ubuntu:~$ tree linux-seguranca-cloud
Command 'tree' not found, but can be installed with:
apt install tree
root@ubuntu:~$ apt install tree
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  squashfs-tools
Use 'apt autoremove' to remove it.
The following NEW packages will be installed:
  tree
0 upgraded, 1 newly installed, 0 to remove and 23 not upgraded.
Need to get 47.4 kB of archives.
After this operation, 111 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 tree amd64 2.1.1-2ubuntu3.24.04.2 [47.4 kB]
Fetched 47.4 kB in 0s (937 kB/s)
Selecting previously unselected package tree.
(Reading database ... 114047 files and directories currently installed.)
Preparing to unpack .../tree-2.1.1-2ubuntu3.24.04.2_amd64.deb ...
Unpacking tree (2.1.1-2ubuntu3.24.04.2) ...
Setting up tree (2.1.1-2ubuntu3.24.04.2) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ubuntu:~$
```

## Tarefa 3 — Executar e registar comandos iniciais

Comandos que permitem identificar o ambiente e consultar informação inicial do sistema.

```
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Setting up tree (2.1.1-2ubuntu3.24.04.2) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...

Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ubuntu:~$ tree linux-seguranca-cloud
linux-seguranca-cloud
├── produto-final
├── topico-01
│   ├── README.md
│   ├── comandos-topico-01.txt
│   └── evidencias
└──

4 directories, 2 files
root@ubuntu:~$ pwd
/root
root@ubuntu:~$ whoami
root
root@ubuntu:~$ hostname
ubuntu
root@ubuntu:~$ uname -a
Linux ubuntu 6.8.0-117-generic #117-Ubuntu SMP PREEMPT_DYNAMIC Tue May  5 19:26:24 UTC 2026 x86_64 x86_64
x86_64 GNU/Linux
root@ubuntu:~$ ls
filesystem  linux-seguranca-cloud
root@ubuntu:~$ date
Fri May 29 14:41:52 UTC 2026
root@ubuntu:~$
```

## Registo dos comandos

```
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GNU nano 7.2  linux-seguranca-cloud/topico-01/comandos-topico-01.txt *
pwd
Mostra o diretorio atual.

whoami
Mostra o utilizador atual.

hostname
Mostra o nome da ma quina.

uname -a
Mostra informacoes do sistema Linux.

ls
Lista ficheiros e diretoios.

date
Mostra a data e hora do sistema

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify
^C Location  ^_ Go To Line ^M-U Undo
^M-E Redo
```

```
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`-- `-- `-- `--
   |-- `-- README.md
   |-- `-- comandos-topico-01.txt
   |-- `-- evidencias

4 directories, 2 files
root@ubuntu:~$ pwd
/root
root@ubuntu:~$ whoami
root
root@ubuntu:~$ hostname
ubuntu
root@ubuntu:~$ uname -a
Linux ubuntu 6.8.0-117-generic #117-Ubuntu SMP PREEMPT_DYNAMIC Tue May  5 19:26:24 UTC 2026 x86_64 x86_64
x86_64 GNU/Linux
root@ubuntu:~$ ls
filesystem  linux-seguranca-cloud
root@ubuntu:~$ date
Fri May 29 14:41:52 UTC 2026
root@ubuntu:~$ nano linux-seguranca-cloud/topico-01/comandos-topico-01.txt
root@ubuntu:~$ cat linux-seguranca-cloud/topico-01/comandos-topico-01.txt
pwd
Mostra o diretorio atual.

whoami
Mostra o utilizador atual.

hostname
Mostra o nome da maquina.

uname -a
Mostra informacoes do sistema Linux.

ls
Lista ficheiros e diretorios.

date
Mostra a data e hora do sistema
root@ubuntu:~$
```

## Tarefa 4 — Criar o primeiro README técnico

```
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root@ubuntu:~$ nano linux-seguranca-cloud/topico-01/README.md
root@ubuntu:~$ nano linux-seguranca-cloud/topico-01/README.md
root@ubuntu:~$ cat linux-seguranca-cloud/topico-01/README.md
# Tópico 1 – Preparacao do ambiente Linux

## Ambiente utilizado
Linux no browser.

## Objetivo desta atividade
Preparar o ambiente Linux e organizar a primeira evidencia tecnica.

## Estrutura criada
Foi criada uma estrutura de diretorios para organizar os ficheiros do modulo.

## Comandos executados
pwd - mostra o diretorio atual
whoami - mostra o utilizador atual
hostname - mostra o nome da maquina
uname -a -mostra informacoes do sistema Linux
ls - lista ficheiros e diretorios
date - mostra a data e hora

## Diferenca entre VM, VPS e infraestrutura em nuvem

VM:
Maquina virtual criada dentro de outro computador.

VPS:
Servidor virtual privado num servidor remoto.

Infraestrutura em nuvem:
Servicos e servidores disponibilizados pela internet.

## Dificuldades encontradas
Alguma dificuldade inicial com comandos Linux.

## Proximos passos
Continuar a praticar comandos e preparar o ambiente para os proximos topicos.
root@ubuntu:~$
```

## Enviar para repositório no github:

```
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Continuar a praticar comandos e preparar o ambiente para os proximos topicos.
root@ubuntu:~$ touch linux-seguranca-cloud/topico-01/ambiente.md
root@ubuntu:~$ nano linux-seguranca-cloud/topico-01/ambiente.md
root@ubuntu:~$ cd linux-seguranca-cloud
root@ubuntu:~/linux-seguranca-cloud$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /root/linux-seguranca-cloud/.git/
root@ubuntu:~/linux-seguranca-cloud$ git commit -m "Primeira atividade Linux"
Author identity unknown

*** Please tell me who you are.

Run

    git config --global user.email "you@example.com"
    git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: unable to auto-detect email address (got 'root@ubuntu.(none)')
root@ubuntu:~/linux-seguranca-cloud$ git config --global user.email "sandypina07@gmail.com"
root@ubuntu:~/linux-seguranca-cloud$ git config --global user.name "sandrafernandes"
root@ubuntu:~/linux-seguranca-cloud$ git remote add origin https://github.com/sandrafernandes/linux-segura
nca-cloud.git
root@ubuntu:~/linux-seguranca-cloud$ git branch -M main
git push -u origin main
error: src refspec main does not match any
error: failed to push some refs to 'https://github.com/sandrafernandes/linux-seguranca-cloud.git'
root@ubuntu:~/linux-seguranca-cloud$ git branch -M main
```

```
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root@ubuntu:~/linux-seguranca-cloud$ git branch -M main
git push -u origin main
error: src refspec main does not match any
error: failed to push some refs to 'https://github.com/sandrafernandes/linux-seguranca-cloud.git'
root@ubuntu:~/linux-seguranca-cloud$ git branch -M main
root@ubuntu:~/linux-seguranca-cloud$ git push -u origin main
error: src refspec main does not match any
error: failed to push some refs to 'https://github.com/sandrafernandes/linux-seguranca-cloud.git'
root@ubuntu:~/linux-seguranca-cloud$ git remote add origin https://github.com/sandrafernandes/linux-segura
nca-cloud.git
error: remote origin already exists.
root@ubuntu:~/linux-seguranca-cloud$ git config --global init.defaultBranch <name>
bash: syntax error near unexpected token `newline'
root@ubuntu:~/linux-seguranca-cloud$ git config --global init.defaultBranch -m
root@ubuntu:~/linux-seguranca-cloud$ git remote add origin https://github.com/sandrafernandes/linux-segura
nca-cloud.git
error: remote origin already exists.
root@ubuntu:~/linux-seguranca-cloud$ git branch -M main
root@ubuntu:~/linux-seguranca-cloud$ git push -u origin main
error: src refspec main does not match any
error: failed to push some refs to 'https://github.com/sandrafernandes/linux-seguranca-cloud.git'
root@ubuntu:~/linux-seguranca-cloud$ git add .
root@ubuntu:~/linux-seguranca-cloud$ git commit -m "Primeira atividade Linux"
[main (root-commit) 5440721] Primeira atividade Linux
 4 files changed, 74 insertions(+)
 create mode 100644 topoico-01/README.md
 create mode 100644 topoico-01/README.txt
 create mode 100644 topoico-01/ambiente.md
 create mode 100644 topoico-01/comandos-topico-01.txt
root@ubuntu:~/linux-seguranca-cloud$ git push -u origin main
Username for 'https://github.com': sandrafernandes
Password for 'https://sandrafernandes@github.com':
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/sandrafernandes/linux-seguranca-cloud.git/'
root@ubuntu:~/linux-seguranca-cloud$ git push -u origin main
Username for 'https://github.com':
Password for 'https://github.com':
remote: No anonymous write access.
fatal: Authentication failed for 'https://github.com/sandrafernandes/linux-seguranca-cloud.git/'
```

```

fatal: Authentication failed for 'https://github.com/sandrafernandes/linux-seguranca-cloud.git/'
root@ubuntu:~/linux-seguranca-cloud$ zip -r linux-seguranca-cloud.zip linux-seguranca-cloud
Command 'zip' not found, but can be installed with:
apt install zip
root@ubuntu:~/linux-seguranca-cloud$ apt install zip
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  squashfs-tools
Use 'apt autoremove' to remove it.
The following NEW packages will be installed:
  zip
0 upgraded, 1 newly installed, 0 to remove and 23 not upgraded.
Need to get 176 kB of archives.
After this operation, 549 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 zip amd64 3.0-13ubuntu0.2 [176 kB]
Fetched 176 kB in 0s (3236 kB/s)
Selecting previously unselected package zip.
(Reading database ... 114054 files and directories currently installed.)
Preparing to unpack .../zip_3.0-13ubuntu0.2_amd64.deb ...
Unpacking zip (3.0-13ubuntu0.2) ...
Setting up zip (3.0-13ubuntu0.2) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ubuntu:~/linux-seguranca-cloud$ zip -r linux-seguranca-cloud.zip .
  adding: topico-01/ (stored 0%)
  adding: topico-01/comandos-topico-01.txt (deflated 41%)

```

```

root@ubuntu:~/linux-seguranca-cloud$ zip -r linux-seguranca-cloud.zip .
  adding: topico-01/ (stored 0%)
  adding: topico-01/comandos-topico-01.txt (deflated 41%)
  adding: topico-01/README.md (stored 0%)
  adding: topico-01/evidencias/ (stored 0%)
  adding: topico-01/README.txt (deflated 49%)
  adding: topico-01/ambiente.md (deflated 34%)
  adding: .git/ (stored 0%)
  adding: .git/HEAD (stored 0%)
  adding: .git/info/ (stored 0%)
  adding: .git/info/exclude (deflated 28%)
  adding: .git/refs/ (stored 0%)
  adding: .git/refs/tags/ (stored 0%)
  adding: .git/refs/heads/ (stored 0%)
  adding: .git/refs/heads/main (stored 0%)
  adding: .git/COMMIT_EDITMSG (stored 0%)
  adding: .git/description (deflated 14%)
  adding: .git/index (deflated 31%)
  adding: .git/logs/ (stored 0%)
  adding: .git/logs/HEAD (deflated 29%)
  adding: .git/logs/refs/ (stored 0%)
  adding: .git/logs/refs/heads/ (stored 0%)
  adding: .git/logs/refs/heads/main (deflated 29%)
  adding: .git/branches/ (stored 0%)
  adding: .git/objects/ (stored 0%)
  adding: .git/objects/64/ (stored 0%)
  adding: .git/objects/64/0e78ab366724053fcbc721771759bf3048bd54 (stored 0%)
  adding: .git/objects/info/ (stored 0%)
  adding: .git/objects/54/ (stored 0%)
  adding: .git/objects/54/407211a480a67f1feab2e8a038cbdde5f4f3d4 (stored 0%)
  adding: .git/objects/9a/ (stored 0%)
  adding: .git/objects/9a/0fb6b303596635b7b971af74c0010326e7e027 (stored 0%)
  adding: .git/objects/pack/ (stored 0%)
  adding: .git/objects/2c/ (stored 0%)
  adding: .git/objects/2c/f36e859bb825a72d525f7b759a6fea011e7603 (stored 0%)
  adding: .git/objects/4b/ (stored 0%)
  adding: .git/objects/4b/825dc642cb6eb9a060e54bf8d69288fbee4904 (stored 0%)
  adding: .git/objects/e6/ (stored 0%)
  adding: .git/objects/e6/9de29bb2d1d6434b8b29ae775ad8c2e48c5391 (stored 0%)

```

```
adding: .git/objects/ (stored 0%)
adding: .git/objects/64/ (stored 0%)
adding: .git/objects/64/0e78ab366724053fcbc721771759bf3048bd54 (stored 0%)
adding: .git/objects/info/ (stored 0%)
adding: .git/objects/54/ (stored 0%)
adding: .git/objects/54/407211a480a67f1feab2e8a038cbdde5f4f3d4 (stored 0%)
adding: .git/objects/9a/ (stored 0%)
adding: .git/objects/9a/0fb6b303596635b7b971af74c0010326e7e027 (stored 0%)
adding: .git/objects/pack/ (stored 0%)
adding: .git/objects/2c/ (stored 0%)
adding: .git/objects/2c/f36e859bb825a72d525f7b759a6fea011e7603 (stored 0%)
adding: .git/objects/4b/ (stored 0%)
adding: .git/objects/4b/825dc642cb6eb9a060e54bf8d69288fbee4904 (stored 0%)
adding: .git/objects/e6/ (stored 0%)
adding: .git/objects/e6/9de29bb2d1d6434b8b29ae775ad8c2e48c5391 (stored 0%)
adding: .git/objects/e4/ (stored 0%)
adding: .git/objects/e4/f2cb60f40eddd3a41d3b861d622b22a1497b1e (stored 0%)
adding: .git/objects/e2/ (stored 0%)
adding: .git/objects/e2/edec1a24c902110c008b59ae8767e669d61139 (stored 0%)
adding: .git/hooks/ (stored 0%)
adding: .git/hooks/sendemail-validate.sample (deflated 58%)
adding: .git/hooks/pre-applypatch.sample (deflated 38%)
adding: .git/hooks/fsmonitor-watchman.sample (deflated 62%)
adding: .git/hooks/pre-receive.sample (deflated 40%)
adding: .git/hooks/post-update.sample (deflated 27%)
adding: .git/hooks/update.sample (deflated 68%)
adding: .git/hooks/pre-merge-commit.sample (deflated 39%)
adding: .git/hooks/commit-msg.sample (deflated 44%)
adding: .git/hooks/pre-rebase.sample (deflated 59%)
adding: .git/hooks/pre-push.sample (deflated 49%)
adding: .git/hooks/applypatch-msg.sample (deflated 42%)
adding: .git/hooks/pre-commit.sample (deflated 45%)
adding: .git/hooks/prepare-commit-msg.sample (deflated 50%)
adding: .git/hooks/push-to-checkout.sample (deflated 55%)
adding: .git/config (deflated 28%)
adding: produto-final/ (stored 0%)
root@ubuntu:~/linux-seguranca-cloud$ ls
linux-seguranca-cloud.zip  produto-final  topico-01
root@ubuntu:~/linux-seguranca-cloud$
```